

EDUCATION

The Chinese University of Hong Kong

B.Sc. in Mathematics

Sophomore · GPA: 3.4/4.0

RESEARCH INTERESTS

- Applied Mathematics
- Computational Imaging
- Inverse Problems
- Statistical Modeling
- Data Science
- Scientific Computing
- Machine Learning

PROJECTS

Image Inverse Problems / Denoising MVP

- Built a small experimental pipeline for image denoising as an entry point to inverse problems.
- Compared Gaussian filtering, Tikhonov regularization, and total variation denoising.
- Evaluated reconstruction quality using PSNR and SSIM.
- Conducted parameter sensitivity analysis and multi-image testing.

Tools: Python, NumPy, pandas, matplotlib, scikit-image

Mathematical Modeling Practice

- Practiced problem formulation, assumption design, model construction, and result interpretation.
- Focused on translating open-ended questions into quantitative models and structured reports.

Tools: Python, LaTeX, Mathematical Modeling

Personal Website

- Built a bilingual personal website for projects, writing, CV, and personal exploration.
- Developed as a static site with reusable content structures and a print-ready CV.

Tools: Astro, TypeScript, HTML, CSS

TECHNICAL SKILLS

Programming: Python, C++, TypeScript basics

Data / Scientific Computing: NumPy, pandas, matplotlib, scikit-image

Web: Astro, HTML, CSS, Vercel

Writing / Tools: LaTeX, Markdown, Git

COURSEWORK

- Calculus
- Linear Algebra
- Probability
- Statistics
- Vector Calculus
- Mathematical Modeling